

Kubernetes Experiment Report Based on Official Tutorials

1. Objective

- Learn how to start a local Kubernetes cluster with Minikube.
- Learn how to deploy, explore, expose, scale, and update an application in Kubernetes.

2. Environment

OS: Windows

Tool: Minikube

Tool: kubectl

Container Runtime: Docker

3.environmental steps

(1) Verified that Minikube, kubectl, and Docker were installed correctly before starting the experiment.

```
C:\Users\31018>minikube version
minikube version: v1.38.1
commit: c93a4cb9311efc66b90d33ea03f75f2c4120e9b0

C:\Users\31018>kubectl version --client
Client Version: v1.29.0
Kustomize Version: v5.0.4-0.20230601165947-6ce0bf390ce3

C:\Users\31018>
C:\Users\31018>docker version
Client:
 Version:           29.3.1
 API version:       1.54
 Go version:        go1.25.8
 Git commit:        c2be9cc
 Built:             Wed Mar 25 16:16:33 2026
 OS/Arch:           windows/amd64
 Context:           desktop-linux

Server: Docker Desktop 4.66.1 (222799)
Engine:
 Version:           29.3.1
 API version:       1.54 (minimum version 1.40)
 Go version:        go1.25.8
 Git commit:        f78c987
 Built:             Wed Mar 25 16:13:48 2026
 OS/Arch:           linux/amd64
```

(2) Started a local Kubernetes cluster with Minikube and verified that the node status was Ready.

Minikube successfully downloaded the required base image and initialized a local Kubernetes cluster.

```

C:\Users\31018>minikube start
* Microsoft Windows 11 Pro 25H2 上的 minikube v1.38.1
* 自动选择 docker 驱动
! Starting v1.39.0, minikube will default to "containerd" container runtime. See #21973 for more info.
* 使用具有 root 权限的 Docker Desktop 驱动程序
* 在集群中 "minikube" 启动节点 "minikube" primary control-plane
* 正在拉取基础镜像 v0.0.50 ...
* 正在下载 Kubernetes v1.35.1 的预加载文件...
  > preloaded-images-k8s-v18-v1...: 272.45 MiB / 272.45 MiB 100.00% 11.08 M
! minikube cannot pull kicbase image from any docker registry, and is trying to download kicbase tarball from github release page via HTTP.
! 您很有可能遇到了网络问题。请确保您至少可以通过 HTTP (直连或代理) 访问互联网。当前您的代理配置为:

  > kicbase-v0.0.50-amd64.tar: 1.28 GiB / 1.28 GiB 100.00% 3.29 MiB p/s 6m3
! minikube was unable to download gcr.io/k8s-minikube/kicbase:v0.0.50, but successfully downloaded kicbase/stable:v0.0.50 as a fallback image
* 创建 docker container (CPU=2, 内存=3800MB) ...|
* 正在 Docker 29.2.1 中准备 Kubernetes v1.35.1--
* 配置 bridge CNI (Container Networking Interface) ...
* 正在验证 Kubernetes 组件...
  - 正在使用镜像 gcr.io/k8s-minikube/storage-provisioner:v5
* 启用插件: storage-provisioner, default-storageclass
* kubectl 未找到。如果你需要使用它, 请尝试: 'minikube kubectl -- get pods -A'
* 完成! kubectl 现在已配置, 默认使用"minikube"集群和"default"命名空间

```

```

C:\Users\31018>kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
minikube      Ready     control-plane   93s   v1.35.1

```

(3) Opened the Kubernetes Dashboard through Minikube to visually inspect cluster resources.

```

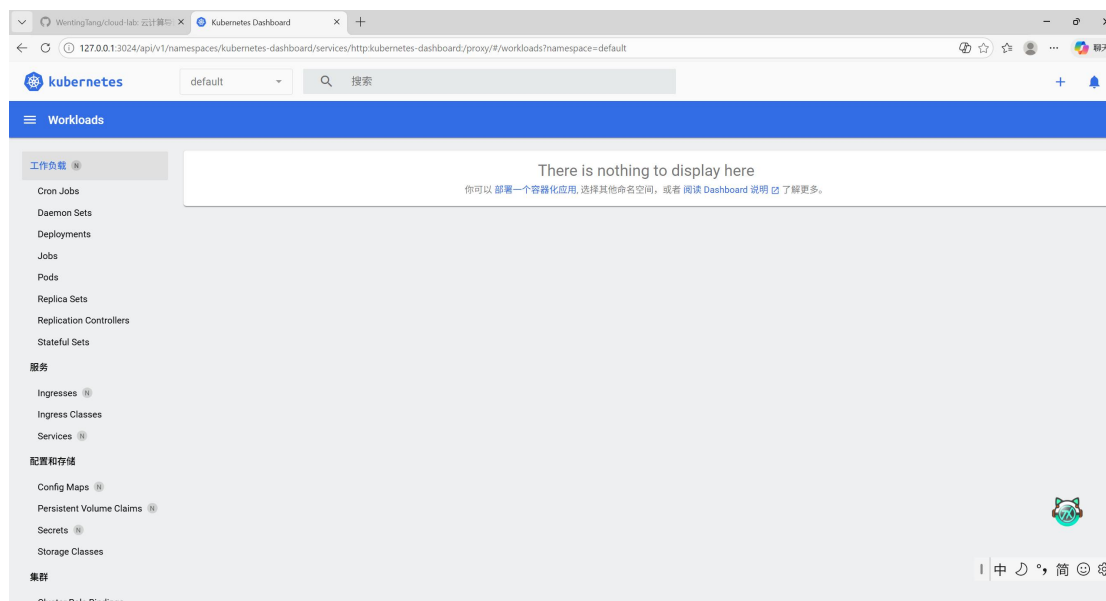
C:\Users\31018>kubectl get nodes
NAME          STATUS    ROLES          AGE   VERSION
minikube      Ready     control-plane   93s   v1.35.1

C:\Users\31018>minikube dashboard
* 正在开启 dashboard ...
  - 正在使用镜像 docker.io/kubernetesui/dashboard:v2.7.0
  - 正在使用镜像 docker.io/kubernetesui/metrics-scraper:v1.0.8
* 某些仪表板功能需要 metrics-server 插件。要启用所有功能, 请运行:

    minikube addons enable metrics-server

* 正在验证 dashboard 运行情况 ...
* 正在启动代理...
  > kubectl.exe.sha256: 64 B / 64 B [-----] 100.00% ? p/s 0s
  > kubectl.exe: 57.32 MiB / 57.32 MiB [-----] 100.00% 856.92 KiB p/s 1m9s
* 正在验证 proxy 运行状况 ...
* 正在使用默认浏览器打开 http://127.0.0.1:3024/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ ...

```



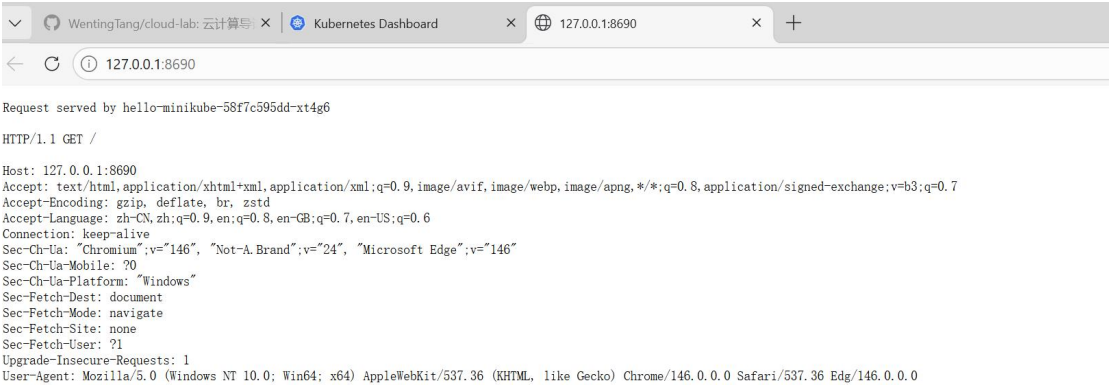
(4) Created a Deployment named hello-minikube and confirmed that the Pod was successfully created and running.

```
C:\Users\31018>kubectl create deployment hello-minikube --image=kicbase/echo-server:1.0
deployment.apps/hello-minikube created

C:\Users\31018>kubectl get deployments
NAME          READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube 1/1     1            1           14s

C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6    1/1     Running   0           20s
```

(5) Exposed the application using a NodePort Service and accessed it successfully through the browser.



```
C:\Users\31018>kubectl expose deployment hello-minikube --type=NodePort --port=8080
service/hello-minikube exposed

C:\Users\31018>kubectl get services
NAME          TYPE        CLUSTER-IP    EXTERNAL-IP    PORT(S)          AGE
hello-minikube NodePort    10.106.55.79  <none>         8080:30631/TCP   7s
kubernetes    ClusterIP   10.96.0.1     <none>         443/TCP          8m55s

C:\Users\31018>minikube service hello-minikube
```

NAMESPACE	NAME	TARGET PORT	URL
default	hello-minikube	8080	http://192.168.49.2:30631

* 为服务 hello-minikube 启动隧道。

NAMESPACE	NAME	TARGET PORT	URL
default	hello-minikube		http://127.0.0.1:8690

* 正通过默认浏览器打开服务 default/hello-minikube...
! 因为你正在使用 windows 上的 Docker 驱动程序, 所以需要打开终端才能运行它。

(6) Checked the application logs using kubectl logs to verify that the container was running correctly.

```
C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6    1/1     Running   0           4m8s

C:\Users\31018>kubectl logs hello-minikube-58f7c595dd-xt4g6
Echo server listening on port 8080.
10.244.0.1:26910 | GET /
10.244.0.1:26910 | GET /favicon.ico
```

Kubernetes Basics — Deploy App

(7) Deployed the sample application kubernetes-bootcamp and verified that the Deployment and Pod were created successfully.

```
C:\Users\31018>kubectl create deployment kubernetes-bootcamp --image=gcr.io/google-samples/kubernetes-bootcamp:v1
deployment.apps/kubernetes-bootcamp created

C:\Users\31018>kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube       1/1     1             1           6m9s
kubernetes-bootcamp  0/1     1             0           8s

C:\Users\31018>kubectl get pods
NAME                                READY   STATUS              RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6    1/1     Running            0           6m16s
kubernetes-bootcamp-67fbdd6b79-dtvj4 0/1     ContainerCreating  0           15s
```

(8) Explored the application resources by inspecting Pods, Nodes, and Deployment details, and entered the container environment with kubectl exec.

```
C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6    1/1     Running   0           7m55s
kubernetes-bootcamp-67fbdd6b79-dtvj4 1/1     Running   0           114s

C:\Users\31018>kubectl get nodes
NAME        STATUS   ROLES    AGE   VERSION
minikube    Ready    control-plane  14m   v1.35.1

C:\Users\31018>kubectl describe pod hello-minikube-58f7c595dd-xt4g6
Name:         hello-minikube-58f7c595dd-xt4g6
Namespace:    default
Priority:      0
Service Account: default
Node:         minikube/192.168.49.2
Start Time:   Mon, 13 Apr 2026 14:36:59 +0800
Labels:       app=hello-minikube
              pod-template-hash=58f7c595dd
Annotations:  <none>
Status:       Running
IP:           10.244.0.5
IPs:          IP: 10.244.0.5
Controlled By: ReplicaSet/hello-minikube-58f7c595dd
Containers:
  echo-server:
    Container ID:  docker://14e2f540ad23c34fe66b41c558d9a63877ba867877b5f66f601e67cd240e7d42
    Image:          kicbase/echo-server:1.0
    Image ID:       docker-pullable://kicbase/echo-server@sha256:127ac38a2bb9537b7f252addff209ea6801edcac8a92c8b1104dadc
```

```
C:\Users\31018>kubectl describe deployment kubernetes-bootcamp
Name:         kubernetes-bootcamp
Namespace:    default
CreationTimestamp: Mon, 13 Apr 2026 14:43:00 +0800
Labels:       app=kubernetes-bootcamp
Annotations:  deployment.kubernetes.io/revision: 1
Selector:     app=kubernetes-bootcamp
Replicas:     1 desired | 1 updated | 1 total | 1 available | 0 unavailable
StrategyType: RollingUpdate
MinReadySeconds: 0
RollingUpdateStrategy: 25% max unavailable, 25% max surge
Pod Template:
  Labels:  app=kubernetes-bootcamp
  Containers:
    kubernetes-bootcamp:
      Image:          gcr.io/google-samples/kubernetes-bootcamp:v1
      Port:           <none>
      Host Port:      <none>
      Environment:    <none>
      Mounts:         <none>
      Volumes:        <none>
  Conditions:
    Type           Status  Reason
    ----           -
    Available       True    MinimumReplicasAvailable
    Progressing     True    NewReplicaSetAvailable
OldReplicaSets:  <none>
NewReplicaSet:   kubernetes-bootcamp-67fbdd6b79 (1/1 replicas created)
Events:
```

Tried to enter the container with kubectl exec, but the image did not include /bin/sh, so interactive shell access was not available.

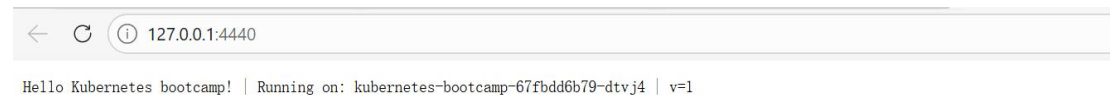
```
C:\Users\31018>kubectl exec -it hello-minikube-58f7c595dd-xt4g6 -- /bin/sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or
directory
command terminated with exit code 127
```

(9) Exposed the Deployment as a Kubernetes Service and confirmed that the application could be accessed externally.

```
C:\Users\31018>kubectl expose deployment kubernetes-bootcamp --type="NodePort" --port 8080
service/kubernetes-bootcamp exposed

C:\Users\31018>kubectl get services
NAME                TYPE        CLUSTER-IP      EXTERNAL-IP  PORT(S)          AGE
hello-minikube      NodePort    10.106.55.79    <none>       8080:30631/TCP   11m
kubernetes          ClusterIP   10.96.0.1       <none>       443/TCP          19m
kubernetes-bootcamp NodePort    10.98.241.185   <none>       8080:31754/TCP   7s

C:\Users\31018>kubectl describe services kubernetes-bootcamp
Warning: v1 Endpoints is deprecated in v1.33+; use discovery.k8s.io/v1 EndpointSlice
Name:                 kubernetes-bootcamp
Namespace:            default
Labels:               app=kubernetes-bootcamp
Annotations:          <none>
Selector:             app=kubernetes-bootcamp
Type:                 NodePort
IP Family Policy:     SingleStack
IP Families:          IPv4
IP:                   10.98.241.185
IPs:                  10.98.241.185
Port:                 <unset> 8080/TCP
TargetPort:           8080/TCP
NodePort:             <unset> 31754/TCP
Endpoints:            10.244.0.6:8080
Session Affinity:     None
External Traffic Policy: Cluster
Events:               <none>
```



(10) Scaled the application from one replica to four replicas and verified that Kubernetes created additional Pods.

```
C:\Users\31018>kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube      1/1     1             1           19m
kubernetes-bootcamp 1/1     1             1           13m

C:\Users\31018>kubectl scale deployments/kubernetes-bootcamp --replicas=4
deployment.apps/kubernetes-bootcamp scaled

C:\Users\31018>kubectl get deployments
NAME                READY   UP-TO-DATE   AVAILABLE   AGE
hello-minikube      1/1     1             1           19m
kubernetes-bootcamp 4/4     4             4           13m

C:\Users\31018>kubectl get pods -o wide
NAME                                READY   STATUS    RESTARTS   AGE   IP            NODE       NOMINATED NODE   READI
NESS GATES
hello-minikube-58f7c595dd-xt4g6     1/1     Running   0          19m   10.244.0.5    minikube   <none>           <none>
>
kubernetes-bootcamp-67fdd6b79-6tkjk 1/1     Running   0          15s   10.244.0.7    minikube   <none>           <none>
>
kubernetes-bootcamp-67fdd6b79-77l5z 1/1     Running   0          15s   10.244.0.8    minikube   <none>           <none>
>
kubernetes-bootcamp-67fdd6b79-bdqsj 1/1     Running   0          15s   10.244.0.9    minikube   <none>           <none>
>
kubernetes-bootcamp-67fdd6b79-dtvj4 1/1     Running   0          13m   10.244.0.6    minikube   <none>           <none>
```

(11) Updated the application image to a new version and observed the rolling update process with kubectl rollout status.

```

C:\Users\31018>kubectl set image deployment/kubernetes-bootcamp kubernetes-bootcamp=nginx
deployment.apps/kubernetes-bootcamp image updated

C:\Users\31018>kubectl rollout status deployment/kubernetes-bootcamp
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 3 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 3 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 3 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 3 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 1 old replicas are pending termination...
deployment "kubernetes-bootcamp" successfully rolled out

C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6     1/1     Running   0           38m
kubernetes-bootcamp-848b67b9b9-24mv7 1/1     Running   0           76s
kubernetes-bootcamp-848b67b9b9-t46gc 1/1     Running   0           2m8s
kubernetes-bootcamp-848b67b9b9-wkh9x 1/1     Running   0           70s
kubernetes-bootcamp-848b67b9b9-zzh85 1/1     Running   0           119s

```

(12) Cleaned up the created Kubernetes resources and stopped the local Minikube cluster after the experiment.

```

C:\Users\31018>kubectl delete service hello-minikube
service "hello-minikube" deleted

C:\Users\31018>kubectl delete service kubernetes-bootcamp
service "kubernetes-bootcamp" deleted

C:\Users\31018>kubectl delete deployment hello-minikube
deployment.apps "hello-minikube" deleted

C:\Users\31018>kubectl delete deployment kubernetes-bootcamp
deployment.apps "kubernetes-bootcamp" deleted

C:\Users\31018>minikube stop
* 正在停止节点 "minikube" ...
* 正在通过 SSH 关闭"minikube"...
* 1 个节点已停止。

```

4.Problems Encountered and Solutions

(1) Slow Image Download During Minikube Startup

When running minikube start, a large base image (such as kicbase) needed to be downloaded, which took a long time. This is because the image must be pulled from a remote registry during the first initialization.

(2) Dashboard Occupying the Terminal

After running minikube dashboard, the terminal was occupied and could not accept further commands. It was later understood that pressing Ctrl + C exits the dashboard proxy without affecting the cluster.

(3) Unable to Access Service via Browser

After obtaining the service URL using minikube service, the browser showed a "connection refused" error. This happened because, on Windows with the Docker driver, the service tunnel must remain running in the terminal; otherwise, the URL becomes invalid.

```
C:\Users\31018>kubectl exec -it hello-minikube-58f7c595dd-xt4g6 -- /bin/sh
OCI runtime exec failed: exec failed: unable to start container process: exec: "/bin/sh": stat /bin/sh: no such file or
directory
command terminated with exit code 127
```

(4) Unable to Enter the Container (kubectl exec Failed)

When executing `kubectl exec -it ... -- /bin/sh`, an error occurred indicating that `/bin/sh` was not found. This is because the container image is minimal and does not include a shell, so interactive access is not available.

(5) Slow Rolling Update Process

When running `kubectl rollout status`, the update process was slow. This is because Kubernetes performs rolling updates, gradually replacing old Pods while waiting for new Pods to become ready.

(6) Rolling Update Failure (Progress Deadline Exceeded)

During the image update, an error “exceeded its progress deadline” occurred. This was caused by failure to pull the new image from `gcr.io`, preventing new Pods from starting. The issue was resolved by rolling back to the previous version.

```
C:\Users\31018>kubectl rollout status deployments/kubernetes-bootcamp
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 4 new replicas have been updated...
Waiting for deployment "kubernetes-bootcamp" rollout to finish: 2 out of 4 new replicas have been updated...
error: deployment "kubernetes-bootcamp" exceeded its progress deadline

C:\Users\31018>kubectl rollout status deployments/kubernetes-bootcamp
error: deployment "kubernetes-bootcamp" exceeded its progress deadline

C:\Users\31018>kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
hello-minikube-58f7c595dd-xt4g6     1/1     Running   0           32m
kubernetes-bootcamp-67fbdd6b79-6tkjk 1/1     Running   0           13m
kubernetes-bootcamp-67fbdd6b79-77l5z 1/1     Running   0           13m
kubernetes-bootcamp-67fbdd6b79-dtvj4 1/1     Running   0           26m
kubernetes-bootcamp-686bd4878-lbzfx 0/1     ImagePullBackOff 0           12m
kubernetes-bootcamp-686bd4878-x82zq 0/1     ImagePullBackOff 0           12m
```

5.conclusion and reflection

Through this experiment, I successfully set up a local Kubernetes cluster using Minikube and learned the basic workflow of Kubernetes, including application deployment (Deployment), resource inspection (Pods and Nodes), service exposure (Service), scaling, and rolling updates. During the experiment, I understood how Kubernetes automatically manages container states through controllers and ensures high availability using replicas. While performing the rolling update, I encountered an image pulling failure, which made me realize the importance of network conditions and image registries in Kubernetes deployments. By troubleshooting the issue and rolling back to the previous version, I gained a deeper understanding of Kubernetes fault-handling mechanisms. Overall, this experiment helped me build an intuitive understanding of Kubernetes and improved my practical skills.